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3 Complex Analysis 2024
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 Mita Ramabulana
Test 2
 9 October 2024



Instructions:

- This question paper consists of **5 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 40 points = 100% (42 points available).

Problem	1	2	3	4	Σ	Mark
Maximum Points	12	8	10	12	40	100%
Result						

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Problem 1

(12 points)

Calculate the following integrals with the help of Cauchy's Integral Theorem, Cauchy's Integral Formula, or Cauchy's Integral Formula for derivatives.

(a) $\int_{\gamma} \frac{\sin z}{z} dz$, where $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$

(b) $\int_{\gamma} \frac{z^2}{z^2 - 3z + 2} dz$, where $\gamma(t) = 5e^{it}$, $t \in [0, 2\pi]$

(c) $\int_{\gamma} \frac{e^z}{(z^2 + 4)^2} dz$, where $\gamma(t) = -i + 2e^{it}$, $t \in [0, 2\pi]$

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Problem 2

(8 points)

- (a) Formulate the Coincidence Principle for functions f and g that are holomorphic in a region $G \subseteq \mathbb{C}$.
- (b) Does there exist a function f that is holomorphic in the open disc $D(0; 1) = \{z \in \mathbb{C} : |z| < 1\}$ such that

$$f\left(\frac{n-1}{2n}\right) = \frac{3n-1}{2n} \quad \text{and} \quad f\left(\frac{n+1}{2n}\right) = \frac{3n+1}{2n} \quad \text{for all } n \in \mathbb{N}?$$

If yes, how many such functions do exist? Justify your answer.

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Problem 3

(10 points)

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a holomorphic function such that $|f(z)| \leq 1$ for all $z \in \mathbb{C}$ with $|z| = 5$. Show that $|f''(w)| \leq \frac{10}{27}$ for all $w \in \mathbb{C}$ with $|w| = 2$.

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Problem 4

(12 points)

- (a) Let f be holomorphic in a punctured disc $D'(z_0; r) = \{z \in \mathbb{C} : 0 < |z - z_0| < r\}$ with some $z_0 \in \mathbb{C}$ and $r > 0$. Define the residue $\text{Res}(f, z_0)$.
- (b) Assume that z_0 is a removable singularity of f . What is $\text{Res}(f, z_0)$? Justify your answer.
- (c) For each of the following functions, determine the type of singularity of f at z_0 . If z_0 is a pole, give its order. Calculate the residue $\text{Res}(f, z_0)$.

(i) $f(z) = \frac{\cos \pi z}{z - \frac{1}{2}}, z_0 = \frac{1}{2}$

(ii) $f(z) = \frac{1}{e^z - 1}, z_0 = 0$